Chapter 10 — Introduction to synthetic data

Chapters 2 through 9 covered how real records are collected, labeled, cleaned, documented, and scaled

Learning objectives

- Scalability benefits that motivate synthetic generation

Definition and analytical value

- Patterns, or structure of real data
- The primary requirement is analytical value
- Synthetic outputs may still miss some real-world noise and edge complexity

Example 10.1 — Cross-Modal Fidelity in Synthetic Data

- Example 10.1 — hands-on module
- Example 10.1 contrasts fidelity across modalities
- Trends seen in real markets
- The same generator cannot be judged by one universal rule; fidelity is modality-specific
- Explore the chapter example module
- View files: `modules/chapter10/example1/`

Generation methods at a glance

- Simulation engines
- Statistical approaches draw from fitted distributions or Monte Carlo models
- Machine learning methods learn an underlying data distribution and emit new samples that
- The method choice depends on modality, privacy goals, and validation capacity

Example 10.2 — Synthetic Patient Records for Privacy

- Example 10.2 — hands-on module
- Example 10.2 shows privacy-preserving medical records
- A healthcare provider can generate synthetic patient tables that mimic real diagnostic and
- This supports research and model development under regulations such as HIPAA and GDPR
- Explore the chapter example module
- View files: `modules/chapter10/example2/`

Cost efficiency and scalability

- Collecting real-world data
- Synthetic generation scales volume on demand and can fill underrepresented classes or edge
- Fraud detection and rare-disease modeling are common cases where synthetic volume helps

Example 10.3 — Synthetic Driving Scenarios for AV Training

- Example 10.3 — hands-on module
- Example 10.3 illustrates synthetic autonomous-driving scenarios
- Simulations can produce varied weather, lighting
- Synthetic clips let perception and planning stacks encounter diverse conditions before
- Explore the chapter example module
- View files: `modules/chapter10/example3/`

Takeaways

- Synthetic data is algorithmically generated yet must preserve analytical structure
- Fidelity requirements differ by modality
- Key benefits include privacy preservation, lower collection cost
- Benefits do not remove the need for validation, which the next part addresses

Next

- Complete the quiz for this part
- Overfitting or lack of diversity when models train only on synthetic copies